PERFECT OBJECTS ADMIT BOUNDED FINITE LOCALLY FREE REPRESENTATIVES UNDER THE RESOLUTION PROPERTY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- A quasi-compact quasi-separated scheme with the resolution property has affine diagonal. invoked at proof lines 9899-9900. Enables the representation proposition.
- On a quasi-compact scheme with affine diagonal every perfect object is represented by a complex of quasi-coherent modules. invoked at proof lines 9901-9903. Produces the representative complex F.
- On a quasi-compact scheme a perfect object is bounded, so a suitable truncation of F is a bounded below complex of quasi-coherent modules representing E. used at proof lines 9904-9907. Prepares the input for the strictly perfect construction.
- A bounded below complex of quasi-coherent modules representing a perfect object on a scheme with the resolution property can be replaced by a strictly perfect complex. invoked at proof lines 9907-9908. Concludes the bounded finite locally free representative.

Source reference. Original lemma 0F8C: sources/stacks-perfect/pinned-original-source.tex, source lines 9705–9705. Cited at proof lines 9899-9900 for the affine diagonal; the representation argument is inside the counted span.

Source reference. Original proposition 08DB: sources/stacks-perfect/pinned-original-source.tex, source lines 1801–1801. Cited at proof lines 9901-9902 to represent E by a complex of quasi-coherent modules; the truncation and conclusion are inside the counted span.

Source reference. Original lemma 0F8E: sources/stacks-perfect/pinned-original-source.tex, source lines 9790–9790. Cited at proof lines 9907-9908 as the final step; the boundedness and truncation arguments are inside the counted span.

Lemma 0.2. *Let X be a quasi-compact and quasi-separated scheme with the resolution property. Then every perfect object of $D(\mathcal{O}_X)$ can be represented by a bounded complex of finite locally free $\mathcal{O}_X$-modules.*

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Proof. Let E be a perfect object of $D(\mathcal{O}_X)$. By Lemma 0F8C we see that X has affine diagonal. Hence by Proposition 08DB we can represent E by a complex $\mathcal{F}^\bullet$ of quasi-coherent $\mathcal{O}_X$-modules. Observe that E is in $D^b(\mathcal{O}_X)$ because X is quasi-compact. Hence $\tau_{\geq n}\mathcal{F}^\bullet$ is a bounded below complex of quasi-coherent $\mathcal{O}_X$-modules which represents E if $n \ll 0$. Thus we may apply Lemma 0F8E to conclude. $\square$